



Does Verbal Street Harassment Signal Perpetrator Dominance to Male and Female Observers?

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Abstract

Objective It is difficult to explain why verbal street harassment, where typically a male harasser yells sexually harassing statements at a female victim, has survived as a behaviour. We propose that verbal street harassment may signal a harasser's dominance and aimed to test this in our registered report.

Methods Participants ($N = 443$) read one of two vignettes describing either a street harassment incident (in which a male perpetrator harasses a female victim) or a street incident without harassment. Participants were asked to evaluate whether the male target possessed a range of traits (including dominance) and to evaluate any harm the female target suffered from the incident.

Results Results suggested that the male target who verbally harassed a female victim on the street was perceived by participants as more dominant and as having a darker personality than a male target who did not engage in street harassment. Participants also perceived the female target as more harmed when she was harassed. However, results did not support the predicted interaction of participant sex and incident type on participants' perceptions of the male and female targets.

Conclusions These results suggest that verbal street harassment may signal a harasser's dominance which may be why the behaviour has been maintained. To establish whether verbal street meets the conditions to be classed as a costly signal, these findings should be replicated and extended. Understanding why street harassment persists as a behaviour is critical to designing effective intervention to prevent street harassment and protect harassment victims.

Keywords Costly signalling theory · Street harassment · Sex differences · Person perception

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Verbal street harassment is commonly experienced by women and is committed most often by a male harasser. In a multi-national survey of street harassment experiences, 57% of women report having experienced harassment (Kearl 2014). Verbal street harassment incidents often involve a male perpetrator making sexualised comments about a female victim's appearance or sexual innuendos (a form of sexual harassment colloquially referred to as "cat-calling"; Frazier et al. 1995). Victims of street harassment experience increased anxiety, feel unsafe in and avoid public spaces to evade harassment (Fairchild and Rudman 2008; Street et al. 2007).

It is difficult to explain why perpetrators engage in verbal street harassment behaviour. Harassment victims tend to ignore harassers to avoid escalating the behaviour (e.g., Gruber 1989). Perpetrators also risk consequences for engaging in street harassment, as harassers can be punished through police orders and fines (e.g., in Australia and the United Kingdom; Walsh and Taylor 2007). Most research to date has examined proximate explanations for harassment behaviour, such as women's reactions to and perceptions of street harassment (e.g., Fileborn and Vera-Gray 2017; Logan 2015; Mellgren et al. 2018). Proximate level explanations do not identify why the behaviour has persisted over time (Scott-Phillips et al. 2011). Understanding why the behaviour survives (i.e., the ultimate explanation for the behaviour) is critical to designing effective interventions to prevent street harassment incidents and protect harassment victims. Costly Signaling Theory offers an ultimate explanation for verbal street harassment behaviour, that harassment behaviour might advertise traits about the harasser.

Costly Signalling Theory

Costly Signalling Theory has been used to explain why human behaviours persist when an immediate benefit does not follow which rewards the behaviour. According to Costly Signalling Theory (Zahavi 1975; Zahavi and Zahavi 1997), costly behaviours can advertise an individual's traits or qualities that are difficult to directly observe. For example, peacocks advertise their genetic quality to peahens by their tail, which is costly to develop and maintain. Large, quality tails make peacocks vulnerable to predators and require significant resources to sustain (e.g., food and energy). However, the development of a costly but high-quality tail means that a peacock's tail is a reliable indicator of genetic quality. Costly signalling theory was originally intended to understand communication between animals. However, recently, the theory has been used to explain human behaviour including purchasing environmentally friendly 'green' products (van den Bergh et al. 2010), posting workout selfies on social media (Vandenbroele et al. 2019) and disruptive classroom behaviours (i.e., students acting as "class clowns"; Wagner 2019).

For example, students who act as "class clowns" disrupt lessons with attempts to make jokes and do so at the learning experience of other students. Compared to students who do not play the "class clown", students who play the "class clown" are more disruptive, aggressive and have lower assessment completion rates (Damico and Purkey 1976). However, despite these negative consequences of the behaviour, relative to their peers, students identified as

“class clowns” have more mutual friends and peer perceived social status (Wagner 2019). In this case, playing the “class clown” signals social affability and pays off for the signaller in the form of a benefit of increased social status, despite the other negative consequences of the behaviour.

We propose that verbal street harassment may function as a costly signal for the harasser. To establish whether a behaviour may be a costly signal, there are several conditions to be met (Bird et al. 2001; Zahavi and Zahavi 1997). First, the behaviour must be clearly observable. Second, the behaviour must be costly to the signaller in some significant domain. Third, the behaviour must be a reliable indicator of an unobservable trait or quality of the signaller. Finally, displaying the behaviour must provide benefit to the signaller. Street harassment is a clearly observable behaviour. There are significant costs associated with the behavior as street harassers risk a range of punishments including police orders to move on (e.g., in Australia and the United Kingdom; Walsh and Taylor 2007), restraining orders (in New Zealand; s9 Harassment Act 1997) or substantial fines (in France and the Netherlands; Boffey 2019; Willsher 2019). Some countries have even criminalized aggressive forms of street harassment (e.g., New Zealand; s8, Harassment Act 1997). However, we do not know what traits verbal street harassment signals about harassers. In this study, we aimed to answer this question by investigating what traits verbal street harassment signals. By examining what verbal street harassment signals, this study takes the next important step in investigating an explanation for why street harassment has survived as a behaviour using the costly signalling theoretical framework.

What Traits Are Signalled by Street Harassment?

Dominance An unobservable trait signalled by street harassment may be dominance. Dominance is an evolved strategy which involves using power and intimidation to gain social ranking in groups (Maner 2017). Social rank is obtained by dominant group members because other group members are fearful of the more dominant group member (de Waal-Andrews et al. 2015). Men are particularly motivated to seek status and are more sensitive to signals of status. For example, male customers spend more money and purchase more expensive products than their female counterparts in the presence of a physically dominant male shop employee (Otterbring et al. 2018). Forcibly imposing sexual attention on women, including instances of sexual assault and sexual harassment, has been theorized to be a means for a perpetrator to signal dominance (Grubbs et al. 2014; Hitlan et al. 2009; Zahavi and Zahavi 1997).

If engaging in verbal street harassment of women signals that the perpetrator is dominant, there are multiple benefits that may follow for the perpetrator. Individuals high in dominance have greater influence over decisions within their group (Cheng et al. 2013). Both in-group and out-group members also perceive individuals high in dominance to have greater influence and status within the group (Cheng et al. 2013). Men who achieve status via dominance also have more allies and higher quality romantic partners (Von Rueden et al. 2010).

Dark Personality Observers often ascribe dark personality traits to perpetrators of crime like psychopathy (a lack of concern for others and social rules; O’Boyle et al. 2012) and

narcissism (an overly high view of self-worth or importance; Morf and Rhodewalt 2001; Edens et al. 2013; Kelley et al. 2019). Moshagen et al. (2018) proposed dark personality traits like psychopathy may be manifestations of a dark personality factor. Individuals who score high on measures of the dark factor of personality are motivated to protect their own interests regardless of whether this causes cost or harm to others (Moshagen et al. 2018). Individuals who measure as high on the dark factor of personality are also more likely to have engaged in criminal or deviant behaviour and measure highly on narcissism and psychopathy (Moshagen et al. 2020). Given that crime perpetrators are perceived as having ‘dark’ personality traits (particularly psychopathy and narcissism; Edens et al. 2013), an anti-social behaviour like verbal street harassment may also signal to others that the perpetrator has a dark personality.

While possessing a dark personality may seem disadvantageous to a harasser, there are potential benefits associated with dark personality traits. For example, narcissists have been shown to be socially popular and are often considered charming at first sight (Back et al. 2010). In this study, we investigated whether street harassment signals that harassers possess traits with potential benefits to the harasser, in particular, dominance and a dark personality.

Who Responds to Verbal Street Harassment?

If verbal street harassment functions as a signal, the behaviour is being maintained by the utility of the behaviour to advertise otherwise unobservable traits like dominance or a dark personality. However, the existing literature on how male and female observers evaluate perpetrators of sexual offences indicates that there could be a sex difference in how male and female observers respond to verbal street harassment. Typically, female observers view perpetrators of sexual offences, like sexual assault and sexual harassment, more negatively than male observers do (Klinke and Meyer 1990; Newcombe et al. 2008). In criminal trials, female observers are more likely to think perpetrators of sexual offences are guilty than male observers (e.g., Schutte and Hosch 1997). Female observers are also more likely to support punishing perpetrators of sexual offences than male observers (van der Bruggen and Grubb 2014; Rotundo et al. 2001) and specifically perpetrators of rape with psychopathy symptoms (Guy and Edens 2003, 2006). These sex difference in how perpetrators of sexual harassment and other sexual offences are perceived have been demonstrated through numerous narrative and meta-analytic reviews of the literature (e.g., Grubb and Harrower 2008; Grubb and Turner 2012; van der Bruggen and Grubb 2014; Rotundo et al. 2001; Suarez and Gadalla 2010; Schutte and Hosch 1997). This suggests there could be differences in the extent to which harasser traits are signaled by verbal street harassment to male and female observers.

Verbal street harassment is anti-social behaviour in which the victim is harmed (e.g., Davidson et al. 2016; Fileborn 2019). As a result of street harassment, victims suffer a range of negative effects including feeling unsafe as well as depression, anxiety and other mental health issues (e.g., Legena 2018; Street et al. 2007). Generally, research suggests observers of sexual harassment, and other sexual offences, perceive the victims of these behaviours have suffered harm or injury as a result (Pickett et al. 2013; Pickel and Gentry 2017). However, female observers consistently perceive

victims of sexual harassment and sexual offences to be more harmed by the offending behavior than male observers (e.g., Grubb and Turner 2012). Female observers also evaluate a broader range of sexual harassment behaviours as threatening, harmful to the victim and warranting punishment of the perpetrator than male observers (e.g., Jensen and Gutek 1982; Kenig and Ryan 1986; Rotundo et al. 2001). We suggest that even though observers may acknowledge the harm caused to the victim by verbal street harassment, the harassment behaviour may still function as signal explaining why the behaviour is maintained despite the anti-social nature of the behaviour. In this study, we measured participants' perceptions of harm to the victim to see if participants view the street harassment incident as harmful, while also responding to verbal street harassment as a signal.

Current Study

Our aim in this study was to assess whether verbal street harassment by a male perpetrator signalled dominance and a dark personality to observers and explore observer sex differences in the traits signalled. In this study, we presented participants with a vignette describing either a typical verbal street harassment incident (in which a male target or perpetrator makes sexualised comments towards a female target or victim; Kearl 2014) or an incident on the street (in which the woman is not harassed). Participants were placed in the role of an observer of the events in the vignette. After reading the vignette, participants were asked about their perceptions of the male target's traits and the harm to the female target from the vignette event.

(H1) Drawing on Costly Signalling Theory we expected an interaction of participant sex and incident type on perceptions of the male target as dominant and possessing a dark personality. Given that men are more sensitive to signals of status (Otterbring et al. 2018) and dominance can lead to status (Maner 2017), we expected male participants would rate the male target (the perpetrator) as more dominant than female participants in the harassment condition but not in the no harassment condition (H1a). Given that female observers are more likely to view perpetrators of sexual offences negatively (e.g., Newcombe et al. 2008; Rotundo et al. 2001) we expected that female participants would rate the male target (perpetrator) as having more dark personality traits than male participants in the harassment condition but not in the no harassment condition (H1b).

(H2) Given that female observers typically perceive sexual offences as more threatening and harmful to victims of sexual violence than men (e.g., Grubb and Harrower 2008; Grubb and Turner 2012; van der Bruggen and Grubb 2014), we also expected an interaction of participant sex and incident type on perceptions of harm to the female target. We expected female participants would perceive the female target (victim) to be more harmed than male participants in the harassment condition but no sex differences in the no harassment condition.

Method

The Stage 1 protocol for this registered report is publicly available at: <https://osf.io/rfwkd>

Participants

Participants were 443 adult (i.e., over 18 years old) American community members recruited online through Amazon Mechanical Turk. Participants were paid USD \$0.55 for their time based on our estimate for the study completion time. This study protocol was approved by the University of Queensland Human Research Ethics Committee (Approved Protocol #2019002887).

Data were analysed using frequentist statistics. We conducted an a priori power analysis via G*Power (Faul et al. 2007) to determine sample size. As the focal effect in this study is novel, it was difficult to select an effect size precedent for power analysis. The effect size of sex differences in how male and female observers perceive perpetrators of sexual offences (including rape, sexual assault and harassment) is consistent and moderate. There have been several meta-analyses of these effects and the effect sizes range from strongly moderate (e.g., $d=0.58$; Suarez and Gadalla 2010) to between small and moderate ($r=0.17$ or $d=0.35$; Schutte and Hosch 1997 or $d=0.42$; Rotundo et al. 2001). A large-scale review of effect sizes across psychology indicates that studies about attitudes have strongly moderate effects (e.g., $r=.27$ or $d=0.56$; Richard et al. 2003). In order to be conservative, we used $d=0.35$ ($f=0.17$) as our effect size precedent. This is the smallest published meta-analytic effect size estimate for sex differences in evaluating perpetrators of sexual offences (Schutte and Hosch 1997). Given our study design, alpha level (0.05) and effect size precedent ($d=0.35$) to achieve 90% power, G*Power indicates 441 participants were required. The G*power output is available with the full study materials at the Open Science Framework: <https://osf.io/sev53/>

Design

In this study, we used a two cell between-subjects experimental design in which the type of street incident was manipulated. There were two conditions, one describing a typical incident of verbal street harassment and one describing a street incident with no harassment. In the harassment condition, a male harasser made sexualized comments about a female victim (based on female victims' accounts of typical verbal street harassment incidents; Kearl 2014; Fairchild 2010; Wesselmann and Kelly 2010). In the no harassment condition, the male and female target had a small interaction on the street which involved no verbal street harassment (Fairchild 2010). We also included participant (i.e., observer) sex as a predictor in our analyses. Our key dependent variables were perceptions of the male target (i.e., harasser) traits and harm to the female target (victim).

Procedure and Materials

Study materials were presented to participants online via Qualtrics. Participants were given an online participant information sheet which detailed what participation involved and their rights. Participants were asked to confirm they consent to participate by clicking the consent button (i.e., participants were asked to provide informed consent). Then participants provided their age, sex, ethnicity and current relationship

status. Participants were randomly allocated to read one of the two vignettes (via a randomizer in Qualtrics) either the harassment or no harassment condition.

After reading the vignette, participants completed a brief questionnaire measuring our key dependent variables. To avoid contamination between the key outcomes, harasser traits were measured first, followed by perceptions of harm to the victim. Each measure of harasser traits and harassment severity were presented on a separate page in Qualtrics and participants were not able to navigate back to measures completed earlier.

Harasser Dominance Perceptions of the harasser's possession of dominance and prestige was assessed using the male peer rating version of the Dominance-Prestige Scale (17 items; Cheng et al. 2010). Participants were asked to rate the harasser (or male target), indicating their agreement to each item on a scale of 1 (*Not at all*) to 7 (*Very much*). The 17-item measure includes eight items measuring dominance ($\alpha = .88$) and nine items measuring prestige ($\alpha = .85$) with acceptable internal consistency for each sub-scale. Factor analyses suggest that the sub-scales are reliably independent (see Cheng et al. 2010). Composite scores for each sub-scale were created by averaging participants scores across items. The dominance sub-scale was used to evaluate H1a and the inclusion of the prestige sub-scale was for exploratory analysis.

Harasser Personality Given that there are strong correlations between individual dark personality traits (e.g., like psychopathy and Machiavellianism or psychopathy and narcissism; Muris et al. 2017; O'Boyle et al. 2012) and strong evidence that the Dark Factor of Personality is the underlying construct beneath these individual traits (Moshagen et al. 2018), we used the 16 item Dark Factor of Personality measure to investigate the extent to which the harasser was perceived as possessing dark personality traits (the D16; $\alpha = .91$; Moshagen et al. 2020). Participants' responses were recorded on a five point Likert scale from 1 (*strongly disagree*) to 5 (*strongly agree*). This measure shows high criterion validity and has high test re-test reliability ($r_{tt} = .90$; Moshagen et al. 2020). In the original inventory, participants were asked to rate themselves on the dark factor of personality traits. To assess perceptions of the male target in our vignettes, we adapted the wording of the items to focus on the harasser in the vignette. For example, the item "It is hard for me to see someone suffering" was adapted to "The man finds it hard to see someone suffering". Items were reverse coded as required and a composite was created for analysis by averaging scores across items contained in the scale. This measure was used to evaluate H1b.

To investigate whether the harasser is seen as possessing other traits, we asked participants to rate the extent to which they found the harasser threatening, masculine, and trustworthy on a seven-point scale from 1 (*Not at all*) to 7 (*Extremely*). These items were drawn from research on person perception (Ma et al. 2015) and were analysed at the item level. These items were included for exploratory analysis.

Harm to the Victim To test H2, we used two measures of harm to the victim (the female target). First, four items were used to assess the harmful effects of the events described in the vignette on the female target. These items were drawn from prior research on street harassment (Fairchild 2010; McCarty et al. 2014; Myers et al. 2002). Three of these items, drawn from McCarty and colleagues work, formed a reliable composite

($\alpha = .89$). If the four items formed a reliable composite (i.e., with a coefficient alpha score of above .80; Bandalos 2018), we planned to analyse these items as a single measure of harm to the victim. The composite score was created by averaging participants' scores across all four items.

The second measure of harm to the victim focused on participants' perceptions of the emotional experience of the female target during the events described in the vignette (a measure used previously by Fairchild 2010). Participants indicated the extent to which they thought the female target experienced eight different emotions, some positive (e.g., excitement) and some negative (e.g., sadness) on a seven-point scale from 1 (*Not at all*) to 7 (*Extremely*). Fairchild (2010) reversed scored the three positive emotions and created a composite averaging across all eight items which showed acceptable reliability ($\alpha = .88$). We compiled data into a composite using Fairchild's (2010) method.

Data Quality Measures

Manipulation Check To ensure that two vignettes were viewed as intended, participants responded to a single item about the extent to which the man verbally harassed the woman on the street. Participants responded on a 7-point scale from 1 (*Not at all*) to 7 (*Very*). We expected the street harassment vignette would be rated as more harassing to the female target than no harassment vignette if the manipulation was successful.

Attention Checks As the experimental manipulation relied on participants reading and attending to the vignette content, we assessed participants' recall of the vignette through a series of attention checks (4 items). First, participants were asked to describe what happened in the vignette they read (as an open-ended free recall question). Then participants were asked to respond to a series of fact-based questions about the content of the vignettes (3 memory recall items). If participants failed to correctly respond to all three recall memory questions and not respond to the free recall question, we planned to exclude their data from the analyses on the basis that they did not attend to the vignette and therefore could not have responded sensibly to the questionnaire.

Demand Characteristic Probe Participants were also asked to respond to two questions about whether there was anything they wanted to tell the experimenters about the study and what they thought the study was about. These questions were intended as a demand characteristic probe (multiple questions are more effective than single questions to detect demand characteristics; Page 1971; Blackhart and Clark 2019). If participants accurately guessed at the experimental hypotheses in response to these questions, then it is possible that they would be subject to demand characteristics (that is, responding in the way they think they should given the study purpose; Kite and Whitley 2018). If participants accurately guessed the experimental hypotheses, we planned to exclude them from the final analysis.

The open-ended questions had a second function, as a recommended bot detection mechanism (see for example Paolacci et al. 2019). There has been discussion about whether some workers (participants) on MTurk might be human assisted bots (and therefore data provided by these participants lacks validity; Matherly 2019). Human assisted bots typically provide nonsensical responses to open-ended questions. By asking participants to respond to three open-ended questions, we were able to manually

screen for responses provided by human assisted bots and remove these prior to data analysis (following guidelines by Paolacci et al. 2019).

After completing the measures, participants were provided with a debriefing information sheet which explained the purposes of the study in full. The debriefing information sheet included contact details for the research team if participants wanted to ask further questions about the study or results.

Analysis Pipeline

Exclusions to Ensure Data Quality As our study relied on vignettes for our experimental manipulation, participants had to pass attention checks so we could verify they had attended to the material in the vignette. Participants data was excluded prior to analysis if they responded incorrectly to all three fact based attentions checks and skipped (i.e., did not respond) the open-ended question asking them to describe the vignette, as this would indicate that they had not attended to the vignette. In addition, we wanted to exclude participants that demonstrated any evidence of demand characteristics, as this could also threaten the integrity of the data. If participants accurately guessed at the experimental hypotheses in their response to the demand characteristic probes, their data was excluded prior to analysis. We planned to continue to recruit participants until we had data from 441 participants which met our data quality criteria as described.

Analytic Strategy for Hypothesis Tests To test H1a and H1b, we planned to conduct two factorial ANOVAs with participant sex and harassment condition entered as independent variables and scores for perceptions of the perpetrator's dominance (via scores on the dominance subscale of the Dominance-Prestige Scale) and dark personality (via scores on the Dark Factor of Personality measure) entered into separate analyses as dependent variables. We planned to follow up significant interactions with contrasts (using Tukey's test) as specified in the hypotheses.

To test H2, we planned to conduct two factorial ANOVAs with participant sex and harassment condition entered as independent variables and scores for the two measures of harm to the victim entered into separate analyses as dependent variables. We planned to follow up significant interactions with contrasts (using Tukey's test) as specified in the hypothesis. An R script for these planned analyses is available on the Open Science Framework at: <https://osf.io/gy2rb/>.

Results

The de-identified dataset is available on Harvard Dataverse at: <https://doi.org/10.7910/DVN/LA3YJK>.

Data Collection and Preparation

When we implemented our data quality exclusion criteria, we realized that we had not included a specific exclusion criteria for our planned bot screening described in the

Stage 1 protocol. We were granted permission by the editor to collect additional data so we could conduct the bot screening as described in the method section of our Stage 1 protocol and meet our target sample size after excluding responses based on the bot screening. We report the planned analyses from the Stage 1 protocol on the data with exclusions as specified by the protocol (as pre-registered; $N = 592$) and with responses that did not pass bot screening excluded (as exploratory; $N = 443$). When collecting our second batch of data via Mturk, we assumed we would see a similar attrition rate for our exclusion criteria and bot screening so we requested a larger batch to account for this attrition. Two more responses than anticipated passed our exclusion criteria and bot screening procedure resulting in 443 valid responses in our final sample.

Only a small proportion of responses (0.2%) had missing data on key outcome variables (see supplementary materials for missing data screening at: <https://osf.io/juqph/>). We used listwise deletion to exclude missing responses from our analyses. Data were prepared and analysed in R (R Core Team 2019).

Sample Characteristics

For our dataset with the pre-registered data exclusion criteria applied, participants were majority male (65.1%). Participants were on average 33.21 years of age ($SD = 10.32$, range = 18–100). Exactly half of participants identified their ethnicity as people of the Americas (50.0%) and a quarter (25.2%) as Southern and Central Asian, with smaller proportions identifying as North-West European (8.1%), South-East Asian (8.1%), North-East Asian (3.2%), Southern and Eastern European (2.9%), Australian (2.4%), Other (1.7%), New Zealander (0.3%), Sub-Saharan African (0.2%) or Oceanian (0.2%). Most participants identified their relationship status as married (58.8%) or single (32.3%) with smaller proportions identifying as in a relationship (8.8%) or divorced (0.2%). After additional exclusions were applied on the basis of bot screening, participant characteristics remained similar (see supplementary materials for a full reporting of participant characteristics for this data).

Overview of Analyses

We report the planned analyses of our data which passed the exclusion criteria specified in our Stage 1 protocol in the pre-registered analyses section below. We replicate our planned analyses on our data which also passed the bot screening procedures in the exploratory analyses section. In the exploratory analyses section, we also report the analyses of the variables we included for exploratory purposes (i.e., harasser traits) and we report Bayes factors for our planned analyses. We calculated Bayes factors so we could quantify the evidence in favour of our experimental hypotheses relative to the null hypothesis (Rouder et al. 2012; Morey et al. 2016). An extended R script for our analyses (including all exploratory analyses) can be found at: <https://osf.io/cj6uk/>

Pre-Registered Analyses

First, we examined the effect of our vignette manipulation of incident type. As expected, participants who read about an incident of street harassment perceived that the man had harassed the woman to a greater extent ($M = 5.90$, $SD = 1.02$) compared to

participants who read the no harassment vignette ($M = 4.83$, $SD = 1.80$), $t(590) = -8.89$, $p < .001$, $d = 0.73$, 95% CI [0.56, 0.90]. The harassment manipulation via the vignettes was effective. A correlation table for all dependent variables is provided in supplementary materials.

To test H1a, concerning the perceived dominance of the male target, we conducted a between-subjects ANOVA as specified in the analysis pipeline. There was a significant main effect of incident type (see Table 1 for ANOVA statistics and reliability coefficients for outcome measures for all pre-registered analyses). Participants' perceived the male target in the harassment condition to be more dominant than participants in the no harassment condition (see Table 2 and Fig. 1 for means, standard deviations and effect sizes for all significant main effects). However, there was no significant main effect of participant sex. We also did not find support for our hypothesized interaction of participant sex and incident type on perceptions of the male target as dominant.

To test H1b, concerning the perceived dark personality of the male target, we conducted a between-subjects ANOVA as specified in the analysis pipeline. Again, there was a significant main effect of incident type, such that the male target in the harassment condition was seen as having a darker personality than the male target in the no harassment condition. There was no significant main effect of participant sex or interaction of participant sex and incident type as predicted in H1b.

To test H2, concerning perceived harm to the female target, we conducted two between-subjects ANOVAs on our two measures of victim harm as specified in the analysis pipeline. The measure which focused on harmful effects of the event to the female target showed acceptable reliability and was analysed as a composite as planned. There was a significant main effect of incident type. Participants who read about the harassment event perceived the female target to be more harmed than those who read about the no harassment event. There was no significant main effect of participant sex or interaction of participant sex and incident type as hypothesized in H2.

For the female target's experience of negative emotions during the incident there was also a significant main effect of incident type. Participants who read about the harassment event perceived the female target to have experienced more intense negative emotions than participants who read about the no harassment event. Again, there

Table 1 ANOVA summary statistics for pre-registered analyses (with pre-registered exclusion criteria implemented)

	Outcome Measures			
	Dominance	Dark Personality	Harm	Emotions experienced
Reliability coefficient (α)	.73	.79	.83	.64
Main effect of incident type	$F(1, 586)=24.08$, $p<.001$	$F(1, 586)=11.66$, $p<.001$	$F(1, 586)=32.40$, $p<.001$	$F(1, 586)=16.54$, $p<.001$
Main effect of participant sex	$F(1, 586)=3.07$, $p=.080$	$F(1, 586)=0.09$, $p=.763$	$F(1, 586)=0.01$, $p=.914$	$F(1, 586)=1.17$, $p=.280$
Participant sex x Incident type	$F(1, 586)=0.12$, $p=.731$	$F(1, 586)=0.06$, $p=.813$	$F(1, 586)<0.01$, $p=.965$	$F(1, 586)=0.01$, $p=.918$

Table 2 Means, standard deviations and effect sizes for main effects of incident type for pre-registered and exploratory analyses on the data (with pre-registered exclusion criteria implemented)

Outcome Measures									
Condition	Target <i>n</i>	Dominance		Dark Personality		Harm		Emotional reaction	
		Male	<i>M (SD)</i>	Female	<i>M (SD)</i>	Male	<i>M (SD)</i>	Masculinity	Prestige
Harassment	296	5.09 (0.80)	3.38 (0.59)	5.50 (0.97)	3.65 (1.22)	5.20 (1.52)	4.93 (1.03)	3.79 (2.25)	5.73 (1.19)
No harassment	296	4.74 (0.80)	3.21 (0.50)	4.93 (1.33)	4.03 (0.91)	5.38 (1.35)	4.60 (0.83)	4.42 (1.90)	5.08 (1.48)
Effect size <i>d</i> [95% CI]		0.44 [0.27, 0.60]	0.31 [0.15, 0.47]	0.49 [0.33, 0.65]	-0.35 [-0.52, -0.19]	-0.13 [-0.29, 0.04]	0.35 [0.19, 0.52]	-0.30 [-0.46, -0.14]	0.48 [0.32, 0.65]

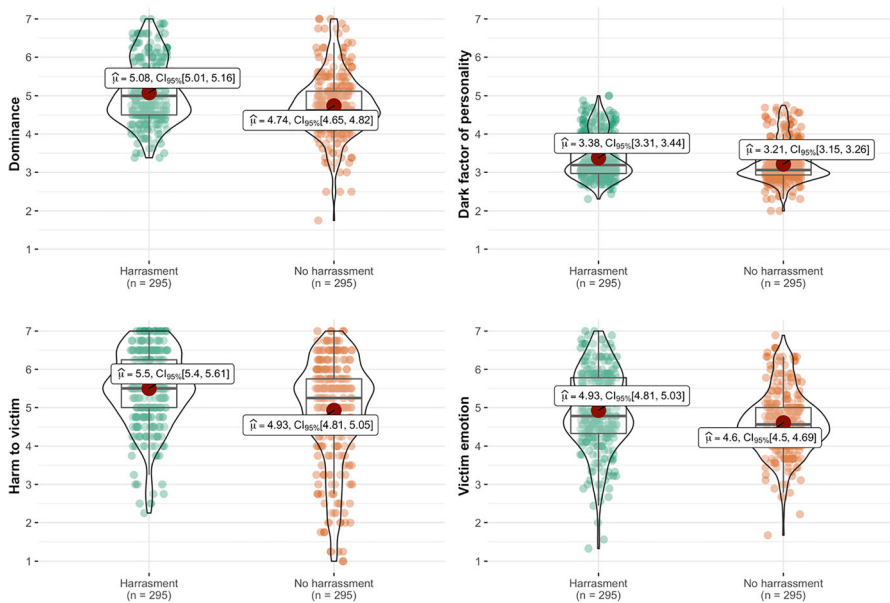


Fig. 1 Violin plots indicating distribution and mean for incident type condition on key dependent variables (with pre-registered exclusion criteria implemented)

was no significant main effect of participant sex or interaction of participant sex and incident type as hypothesized in H2. Collectively across the key perceptions of the male and female target, there was no support for our hypothesized interactions of participant sex and incident type.

Exploratory Analyses

Analysis of Exploratory Outcome Measures We asked participants about their perceptions of the male target's prestige, masculinity, trustworthiness and how threatening the male target was. We conducted four between-subjects ANOVAs with participant sex and incident type entered as independent variables and each trait construct entered separately as an outcome variable. For the analysis of the male target's perceived masculinity, there were no significant main effects of participant sex or incident type or interaction of participant sex and incident type (see Table 3 for ANOVA statistics). For the analyses of the male target's perceived prestige, trustworthiness and extent of threat, there were significant main effects of incident type but no significant main effects of participant sex or interactions of participant sex and incident type (see Table 3). The male target in the harassment condition was perceived to have less prestige, as being less trustworthy and more threatening than the male target in the no harassment condition (see Table 2 for means, standard deviations and effect sizes for significant main effects).

Re-Analysis of Data after Bot Screening We repeated our planned analyses and analyses of exploratory variables on our dataset after the responses that did not pass bot screening were also excluded. The vignette manipulation of incident type remained

Table 3 ANOVA statistics for analysis of exploratory variables (with pre-registered exclusion criteria implemented)

	Outcome Measures			
	Masculinity	Prestige	Trustworthy	Threatening
Reliability coefficient (α)	–	.81	–	–
Main effect of incident type	$F(1, 586)=1.45$, $p=.229$	$F(1, 586)=17.91$, $p<.001$	$F(1, 586)=12.53$, $p<.001$	$F(1, 586)=34.42$, $p<.001$
Main effect of participant sex	$F(1, 586)=0.04$, $p=.844$	$F(1, 586)=0.30$, $p=.586$	$F(1, 586)=0.10$, $p=.753$	$F(1, 586)=0.72$, $p=.395$
Participant sex x Incident type	$F(1, 586)=0.47$, $p=0.49$	$F(1, 586)=0.26$, $p=.609$	$F(1, 586)=0.30$, $p=.583$	$F(1, 586)=0.83$, $p=.363$

effective. Participants in the harassment condition perceived the male target to have harassed the female target to a greater extent ($M = 6.02$, $SD = 0.96$) than participants in the no harassment condition ($M = 4.49$, $SD = 1.90$), $t(441) = 10.59$, $p < .001$, $d = 1.03$, 95% CI [0.83, 1.22]. A correlation table for all dependent variables is provided in supplementary materials. Overall, the pattern of results remained the same for analyses of key and exploratory outcome variables. Importantly, reliability coefficients for the outcome measures and effect size estimates for the main effects of incident type on the outcome variables all strengthened after responses which failed bot screening were excluded.

There were significant main effects of incident type on participants' perceptions of the male target as dominant and having a dark personality. There were also significant main effects of incident type on participants' perceptions of the harm caused to and negative emotions experienced by the female target (see Table 4 for the ANOVA statistics for these analyses). Consistent with the results of the planned analyses on the pre-registered data exclusions, participants who read about the harassment incident perceived the male target to be more dominant and as having a darker personality than those who read about the no harassment incident (see Table 5 and Fig. 2 for means, standard deviations and effect sizes). Participants who read about the harassment

Table 4 ANOVA statistics for analysis of key variables (after responses excluded through bot screening)

	Outcome Measures			
	Dominance	Dark Personality	Harm	Emotions experienced
Reliability coefficient (α)	.73	.82	.84	.70
Main effect of incident type	$F(1, 439)=25.46$, $p<.001$	$F(1, 439)=11.74$, $p<.001$	$F(1, 439)=50.17$, $p<.001$	$F(1, 439)=23.08$, $p<.001$
Main effect of participant sex	$F(1, 439)=1.72$, $p=.189$	$F(1, 439)<0.01$, $p=.951$	$F(1, 439)=0.27$, $p=.602$	$F(1, 439)=1.13$, $p=.288$
Participant sex x Incident type	$F(2, 439)=0.25$, $p=.614$	$F(1, 439)<0.01$, $p=.897$	$F(1, 439)=0.77$, $p=.382$	$F(1, 439)=0.63$, $p=.428$

Table 5 Means, standard deviations and effect sizes for main effects of incident type for pre-registered and exploratory analyses on the data (with bot screening exclusions implemented)

Outcome Measures													
		Target	Dominance		Dark Personality		Harm	Emotional reaction		Masculinity	Prestige	Trustworthiness	Threatening
			Male				Female						
			<i>M (SD)</i>		<i>M (SD)</i>		<i>M (SD)</i>		<i>M (SD)</i>		<i>M (SD)</i>		<i>M (SD)</i>
Condition		<i>n</i>											
Harassment		228	5.19 (0.84)		3.49 (0.62)		5.56 (1.00)		5.11 (1.06)		3.41 (1.28)		5.77 (1.25)
No harassment		215	4.76 (0.90)		3.28 (0.57)		4.72 (1.43)		4.66 (0.92)		3.87 (1.00)		4.92 (1.58)
Effect size (<i>d</i>)			0.50		0.34		0.68		0.45		-0.40		0.60
[95% CI]			[0.31, 0.68]		[0.15, 0.52]		[0.49, 0.88]		[0.26, 0.64]		[-0.59, -0.21]		[0.41, 0.79]

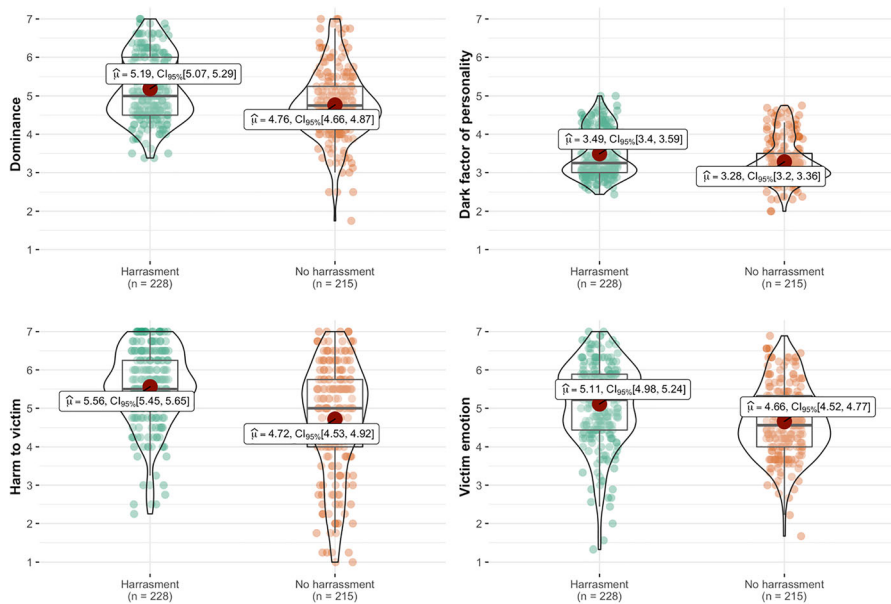


Fig. 2 Violin plots indicating distribution and mean by incident type condition on key dependent variables (with bot screening exclusions implemented)

incident also rated the female target as more harmed and as experiencing more negative emotions during the incident than those who read about the no harassment incident (see Table 5). There were no significant main effects of participant sex or interactions of participant sex and incident type on these key variables. These analyses did not provide any support for the interactions expected in H1a, H1b and H2.

There were also significant main effects of incident type on participants' perceptions of the male target as having prestige, being trustworthy and threatening (see Table 6 for ANOVA statistics). Consistent with the earlier analysis of the exploratory variables, participants who read about the harassment incident perceived the male target to have less prestige, as less trustworthy and more threatening than participants who read about

Table 6 ANOVA statistics for analysis of exploratory variables (after responses excluded through bot screening)

	Outcome Measures			
	Masculinity	Prestige	Trustworthy	Threatening
Reliability coefficient (α)	—	.84	—	—
Main effect of incident type	$F(1, 439)=0.07$, $p=.790$	$F(1, 439)=15.57$, $p<.001$	$F(1, 439)=11.99$, $p<.001$	$F(1, 439)=40.48$, $p<.001$
Main effect of participant sex	$F(1, 439)<0.01$, $p=.936$	$F(1, 439)=0.99$, $p=.321$	$F(1, 439)=0.04$, $p=.846$	$F(1, 439)=0.12$, $p=.724$
Participant sex x Incident type	$F(1, 439)=1.74$, $p=.187$	$F(1, 439)<0.01$, $p=.999$	$F(1, 439)=0.04$, $p=.845$	$F(1, 439)=1.42$, $p=.235$

the no harassment incident (see Table 5 for means, standard deviations and effect sizes). Again, there was no significant main effect of incident type on participants' perceptions of the male target as masculine. There were also no significant main effects of participant sex or interactions of participant sex and incident type on participants' perceptions of the male target as having prestige, being trustworthy, threatening or masculine.

Bayesian Analyses We calculated Bayes factors to quantify the evidence in favour of the null and alternative hypotheses (BF_{10}) for our planned analyses testing our experimental hypotheses. Although Bayes factors provide a continuously scaled evaluation of the strength of evidence in favour of either the null or alternate hypothesis, the evidence can be categorized as moderate, strong and extreme using categories (Lee and Wagenmakers 2014). For Bayes factors testing evidence for the alternate hypothesis (BF_{10}), factors between 3 and 10 provide moderate evidence for the alternate hypothesis, greater than 10 provides strong evidence for the alternate hypothesis and greater than 100 provides extreme evidence of the alternate hypothesis (Lee and Wagenmakers 2014). Bayes factors between 0.33–0.1 provide moderate evidence for the null hypothesis, less than 0.1 provides strong evidence for the null hypothesis and less than 0.01 provides extreme evidence for the null hypothesis. To calculate Bayes factors, we used objective default priors via the BayesFactor package in R. We used default priors as they are a conservative test of the null hypothesis (Rouder et al. 2012).

For the key outcome measures, perceptions of the male target as dominant and possessing a dark personality and perceptions of the female target's emotions and harm experienced, the Bayes factors corroborated the results of the frequentist analyses. We calculated Bayes factors examining whether each effect (e.g., the main effects and interaction) in isolation were better explanations for the data than the null hypotheses. Consistently, the main effect of incident type had a Bayes factor indicating there was strong to extreme evidence that the data were more likely under the alternate than the null hypothesis (see Table 7 for Bayes factors). The Bayes factors for the main effects of participant sex and the interaction of participant sex and incident type consistently indicated moderate to strong evidence that the data were more likely under the null hypothesis than the alternate hypothesis (see Table 7).

Discussion

Our first aim was to investigate how observers perceived perpetrators of street harassment, in particular whether perpetrators were viewed as more dominant and as having a dark personality, and whether there was participant (i.e., observer) sex differences in these perceptions. We hypothesized that male participants would rate the male target as more dominant than female participants in the harassment condition but not in the no harassment condition (H1a). Our results suggested that male targets who engaged in street harassment were perceived as more dominant than those who did not engage in street harassment. However, there was no support for our hypothesized interaction of observer sex and incident type via planned frequentist or exploratory Bayesian

Table 7 Bayes factors (BF_{10}) for analyses of key outcome measures

Data set	Outcome Measures							
	Dominance		Dark Personality		Harm		Emotions experienced	
	Pre-registered data exclusions	After bots screening	Pre-registered data exclusions	After bots screening	Pre-registered data exclusions	After bots screening	Pre-registered data exclusions	After bots screening
Main effect of incident type	41,008.95	26,992.13	58.81	72.26	2,067,126.00	3,328,960,540.00	477.94	5454.31
Main effect of participant sex	0.34	0.19	0.10	0.11	0.10	0.15	0.15	0.15
Participant sex x Incident type	0.52	0.23	0.21	0.25	0.30	0.38	0.18	0.16

analyses. We also expected female participants would perceive the male target as having a darker personality than male participants in the harassment condition but not in the no harassment condition (H1b). Our results did not support the hypothesized interaction of observer sex and incident type. Again, regardless of sex, observers evaluated the male target who engaged in street harassment as having a darker personality compared to the male target who did not engage in street harassment.

A second aim of this study was to investigate whether there were observer sex differences in how participants perceived victims of street harassment to be harmed by the incident. We expected that in the harassment condition female participants would evaluate the female target as more harmed than male participants but anticipated no effect of participant sex in the no harassment condition (H2). We found no support for our hypothesized interaction of participant sex and incident type on perceptions of harm to the female target via planned frequentist or exploratory Bayesian analyses. Regardless of sex, observers viewed the female target to be more harmed and to have experienced more negative emotions in the harassment condition relative to the no harassment condition.

We were interested in whether street harassment might signal other perpetrator traits to observers, like prestige, masculinity, trustworthiness or being threatening and we included measures of these traits for exploratory purposes. Our exploratory analyses suggest that participants perceived the male target who engaged in street harassment as having less prestige, being more threatening and less trustworthy. Overall, street harassment signals several traits about a harassment perpetrator, even while observers recognize that the harassment behavior is harmful to the harassment victim.

Street Harassment as a Costly Signal

Our study provides preliminary evidence that verbal street harassment signals several traits about harassment perpetrators including dominance, dark personality, prestige, threat and lack of trustworthiness. Despite the behaviour being punishable, harmful, and without any obvious immediate reinforcement, signalling could explain why the behaviour has persisted over time. Although the traits signaled by street harassment may not seem advantageous, there are potential fitness benefits to these traits. Signalling dominance or dark personality offers several benefits. For example, individuals with greater dominance exert more influence over decisions in social groups, have greater status and more valuable social connections (Cheng et al. 2013; Von Rueden et al. 2010). Similarly, individuals with dark personalities are often socially popular (Back et al. 2010) and able to access resources through their social networks (e.g. Mogilski and Welling 2017) which may bestow a range of benefits to these individuals.

Using vignettes as a preliminary test of the traits perpetrators signal by engaging in street harassment is a limitation of our study. While we found that verbal street harassers are perceived to be dominant and have dark personality traits, it is unclear whether harassment perpetrators actually possess these traits. In addition, our reliability coefficients for the measures used for dominance, prestige and dark personality were lower than that reported in the validation papers for these measures (see Cheng et al. 2010; Moshagen et al. 2020). This may be a function of the changes we made to these measures so they could be used by participants to evaluate the male target in our vignettes. The male target in our vignettes was unfamiliar to the participants and both

the Dominance-Prestige Scale and Dark Factor of Personality scale are designed to evaluate well known targets (in the validated versions of these scales participants either rate a peer or themselves). These limitations could be addressed in future field study. Researchers could focus on investigating traits possessed by street harassers by surveying male street harassers and men who do not engage in street harassment to examine how dominant or dark their personalities are. This would also allow the Dominance-Prestige Scale and Dark Factor of Personality scale to be used in their validated format (in which participants would rate themselves).

Although the harassment manipulation was successful, participants still perceived the male target to have harassed the female target in the control condition (scores for the control condition on our manipulation check were above the scale mid-point). Participants also viewed the female target in the control condition to be harmed by the incident (scores in the control condition for both measures of victim harm were also above the mid-point). We kept the setting of the control vignette the same as the harassment vignette (i.e., the female target was alone, the event took place at night). Even though the male target does not directly harass the female target, the female target is alone near a group of men behaving in a rowdy manner, behaving in a way that may suggest the men were intoxicated. Participants may have perceived the control vignette to be harassing to the woman because the situation could escalate to harassment or assault or because participants believed the female target felt threatened even though there was no direct harassment (above mid-point scores on the victim harm measures for the control condition support this). This is also consistent with research indicating that women are more fearful of 'stranger' sexual assaults (e.g. Hickman and Muehlenhard 1997) and perceivers over-estimate the frequency of 'stranger' sexual assaults (e.g., Sleath and Woodhams 2014; which take place in settings similar to our control and harassment vignettes). Because participants also viewed our control vignette as harassing for the female target, our results may underestimate the magnitude of the effect of street harassment as a signal of dominance. Future research could use a control vignette which focuses on a social interaction set during day time to investigate whether our data underestimates the size of observed effects.

Observer Sex Differences

Although we did not find support for the hypothesized participant sex differences, our results have implications for the broader literature on how observers evaluate victims and perpetrators of sexual harassment and other sexual offences. There have been mixed findings in respect of whether there are observer sex differences in how victims and perpetrators of sexual harassment are perceived. However, several reviews of this literature report an observer sex effect. Generally, female observers evaluate perpetrators of sexual harassment and sexual offences more negatively and as more deserving of punishment than male observers (Rotundo et al. 2001; Schutte and Hosch 1997; van der Bruggen and Grubb 2014). Female observers also typically perceive victims of sexual harassment and offences more positively than male observers (e.g. Grubb and Turner 2012; Rotundo et al. 2001). We found no significant effect of observer sex on perceptions of the perpetrator's traits or the harm suffered by the victim. It is also important to note that although we conducted a priori power analyses to determine our sample size, the minimum effect we were powered to detect was a small to moderate effect. Our analyses

would not be able to reliably detect any effect that was smaller than this. However, in our exploratory Bayesian analyses, there was moderate to strong evidence for the null hypothesis as an explanation for our data rather than the alternate hypothesis that there was a sex difference in participants' perceptions of the perpetrator and victim (either directly or via an interaction with incident type). Our results are consistent with a more recent review which suggests there may not be a difference in how male and female observers perceive victims of sexual assault (Gravelin et al. 2019).

It is possible that the lack of a sex difference in our results may indicate that there has been a shift in attitudes about the acceptability of sexual harassment. Some scholars have argued that observer sex differences towards sexual harassment and sexual offences can be accounted for by underlying differences in attitudes towards sexual offending. For example, men more strongly endorse rape myths (misleading beliefs about how sexual assaults occur; Anderson et al. 1997; Suarez and Gadalla 2010) and some research suggests this is what leads them to view victims more negatively and perpetrators more positively than female observers do (e.g. Lee et al. 2010). Rape myth endorsement, interacting with observer sex, has also been found to effect how people perceive sexual assault victims (Bohner and Schapansky 2018). There is emerging evidence that other attitudes about sexual behaviour mediate the relationship between observer sex and how sexual assault victims are perceived (e.g. Angelone et al. 2020; Lynch et al. 2020). In our study, we did not assess participants attitudes towards the acceptability of sexual harassment, or other related attitudes like rape myth endorsement, so we cannot establish whether this may explain why we did not find evidence of a participant sex effect on how the male and female target were perceived. This question could be addressed in future research on verbal street harassment.

Implications and Conclusion

To date, limited research has been conducted on street harassment and even less research has focused on why perpetrators engage in street harassment. Although we found preliminary evidence that street harassment signals dominance and dark personality traits about a harassment perpetrator, these findings need to be replicated and extended to confirm that verbal street harassment meets the other conditions for the behaviour to qualify as a signal (Bird et al. 2001; Zahavi and Zahavi 1997). Understanding why street harassment persists is important to the development of effective primary prevention to protect harassment victims from the negative effects of harassment on mental health and sense of safety (Davidson et al. 2016; Legena 2018; Street et al. 2007). Importantly, if verbal street harassment is a costly signal, this would suggest that punishments (like fines or criminal charges) may not be an effective deterrent to prevent perpetrators from engaging in street harassment. To effectively deter perpetrators, it may be more productive to try and target the inferences that observers make about harassment perpetrators in order to reduce the impact of verbal street harassment as a signal. It may be possible to achieve this through education campaigns which explain how harassment perpetrators may be benefitting from verbal street harassment in order to reduce the extent to which observers ascribe traits to harassment perpetrators.

Our study considered how third-party observers perceive a common form of verbal street harassment in which a male perpetrator engages in sexual harassment of a female victim. Third party observers, who are not involved in the incident, may perceive street harassment differently to observers who have a relationship with the perpetrator (e.g. Barnett et al. 1992; Osman 2016). Some research on street harassment suggests that male harassers are more likely to engage in street harassment when with a group of their peers (Quinn 2002; Wesselmann and Kelly 2010). It is possible that having a relationship with the perpetrator (or victim) of a street harassment incident may change how observers perceive both the perpetrator and the victim. Future research should also investigate whether a relationship between the perpetrator or victim and the observer changes the function of verbal street harassment as a costly signal. This future research may also result in productive avenues to develop primary prevention strategies for different groups of observers in order to reduce street harassment incidents and protect harassment victims.

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Declarations

Ethics This protocol was cleared by the University of Queensland Human Research Ethics Committee (Approved Protocol #2019002887).

Conflict of Interest The authors declare that they have no conflict of interest.

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